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Task sheets on conditional statements

1. Write a program that checks if a given number is positive. If it is positive, print "Number is positive"; otherwise, print "Number is not positive."
2. Write a program that checks if a given number is even. If it is even, print "Number is even"; otherwise, print "Number is odd."
3. Write a program that checks if a given number is divisible by both 2 and 3. If it is, print "Number is divisible by 2 and 3"; otherwise, print "Number is not divisible by both 2 and 3."
4. Write a program that checks if a given year is a leap year. If it is a leap year, print "Year is a leap year"; otherwise, print "Year is not a leap year." (Hint: A leap year is divisible by 4 but not by 100, except if it is divisible by 400.)
5. Write a program that calculates the grade based on a student's score. If the score is above 90, print "Grade: A"; if it is between 80 and 90, print "Grade: B"; if it is between 70 and 80, print "Grade: C"; if it is between 60 and 70, print "Grade: D"; otherwise, print "Grade: F."
6. Write a program that checks if a given string, is a palindrome. If it is, print "String is a palindrome"; otherwise, print "String is not a palindrome." (Hint: A palindrome is a word, phrase, number, or other sequence of characters that reads the same forward and backward.)
7. Write a program that checks if a given character, is a vowel. If it is a vowel, print "Character is a vowel"; otherwise, print "Character is not a vowel."
8. Write a program that determines the largest of three given numbers. Print the largest number.

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9. Write a program that checks if a given number, is a prime number. If it is a prime number, print "Number is prime"; otherwise, print "Number is not prime." (Hint: A prime number is a number greater than 1 that is divisible only by 1 and itself.)
10. Write a program that determines the number of days in a given month. The program should ask the user to input the month number (1 for January, 2 for February, etc.) and print the number of days in that month. If an invalid month number is entered, print "Invalid month number."
11. Write a program that determines the cost of a movie ticket based on the age of the person. If the age is 12 or below, the ticket cost is \$5; if the age is between 13 and 17 (inclusive), the ticket cost is \$7; if the age is 18 or above, the ticket cost is \$10.
12. Write a program that determines the season based on a given month. If the month is December, January, or February, print "Winter"; if it is March, April, or May, print "Spring"; if it is June, July, or August, print "Summer"; if it is September, October, or November, print "Autumn."
13. Write a program that determines the discount amount based on a customer's membership level. If the level is "Gold", apply a 20% discount; if it is "Silver", apply a 15% discount; if it is "Bronze", apply a 10% discount; otherwise, no discount is applied.
14. Write a program that simulates a vending machine. Ask the user to enter the amount of money they have, and display a menu of items with their prices. Prompt the user to select an item by entering its corresponding number. If the user has enough money, deduct the price from their balance and display the item's name along with their remaining balance. If the user does not have enough money, display a message indicating the insufficient funds.
15. Write a program that determines a person's BMI (Body Mass Index) category based on their weight and height. Ask the user to enter their weight in kilograms and their height in meters. Calculate their BMI using the formula: $BMI = \text{weight} / (\text{height} * \text{height})$

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height). Display a message indicating their BMI category based on the following ranges: "Underweight" for BMI less than 18.5, "Normal weight" for BMI between 18.5 and 24.9, "Overweight" for BMI between 25 and 29.9, and "Obese" for BMI greater than or equal to 30.

16. Write a program that assists a customer in selecting a suitable mobile phone plan. Ask the user to enter the number of minutes they anticipate using per month and display the plan options: "Basic," "Standard," and "Premium." If the user enters less than 200 minutes, recommend the "Basic" plan. For 200-500 minutes, recommend the "Standard" plan. For more than 500 minutes, recommend the "Premium" plan.
17. Write a program that calculates the shipping cost based on the weight and destination of a package. Ask the user to enter the weight in kilograms and the destination: "Domestic" or "International." For domestic shipping, charge \$2 per kilogram. For international shipping, charge \$5 per kilogram. Display the total shipping cost.
18. Write a program that determines the eligibility of a person to apply for a driver's license based on their age and the type of license. Ask the user to enter their age and the type of license they are applying for: "Car" or "Motorcycle." If the age is 18 or above for a car license or 16 or above for a motorcycle license, print "Eligible to apply for a [license type] license"; otherwise, print "Not eligible to apply for a [license type] license."
19. Write a program that helps a user choose the right clothing for the weather. Ask the user to enter the current temperature in Celsius. Based on the temperature, provide suggestions for clothing: if the temperature is below 10 degrees, suggest "Winter jacket, hat, and gloves"; if it is between 10 and 20 degrees, suggest "Sweater or light jacket"; if it is above 20 degrees, suggest "T-shirt and shorts."

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20. Write a program that determines the discount amount for a customer's online purchase. Ask the user to enter the total order amount. If the order amount is above \$100, apply a 20% discount; if it is between \$50 and \$100, apply a 10% discount; if it is below \$50, apply a 5% discount. Print the discounted amount.

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